

From mining void to resource: Water quality of coal mine pit lakes in South Kalimantan

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Abstract: Coal mining in South Kalimantan has created numerous pit lakes that local communities use as water sources, although their quality has not been comprehensively assessed. This study presents an integrated evaluation of pit lake water quality by combining regulatory compliance with STORET standards (STORage and RETrieval), contamination factor (CF), and pollution load index (PLI), while also examining the effect of water depth on quality variation. Water samples were collected using grab sampling and analysed for temperature, total suspended solids (TSS), total dissolved solids (TDS), pH, biological oxygen demand (BOD), chemical oxygen demand (COD), dissolved oxygen (DO), Fe, Pb, and Mn in accordance with Government Regulation of the Republic of Indonesia Number 22 of 2021. Differences between surface water and 5 m depth were tested using the Independent Samples *t*-test and Mann–Whitney *U* test. The results showed that BOD indicated moderate contamination ($CF = 2.68$), whereas the other parameters were in the low contamination category ($CF < 1$). The overall PLI was 0.348, indicating that the water had acceptable quality. Water depth up to 5 m did not significantly affect most parameters, although pH at 5 m was slightly lower than at the surface. These findings indicate that pit lake water requires treatment before domestic use and support the need for regular monitoring and sustainable management.

Keywords: biological oxygen demand (BOD), coal mining, contamination factor, pit lake, pollution load index, water treatment

INTRODUCTION

Open-pit coal mining commonly leaves unreclaimed voids that fill with rainwater and groundwater, forming pit lakes. These artificial lakes may represent both an environmental liability and a valuable water resource which is highly relevant in Mekar Jaya Village, Angsana District, South Kalimantan. There a former coal mine pit lake covering approximately 350,000 m² is already used for bathing, washing, sanitation, agriculture, livestock, and fisheries (Denimal *et al.*, 2005; Fuentes-López *et al.*, 2022; Jiang *et al.*, 2023; Li *et al.*, 2023; Cooke, Emmerton and Drevnick, 2024; Sekudewicz *et al.*, 2024; Batista-Cruz *et al.*, 2025; Cooke *et al.*, 2025; Iranmanesh *et al.*, 2025). When poorly managed, these water bodies may generate acid mine drainage and mobilise

contaminants, posing risks to surface water, groundwater, and surrounding ecosystems (Fuentes-López *et al.*, 2022; Cooke, Emmerton and Drevnick, 2024; Batista-Cruz *et al.*, 2025). In contrast, pit lakes may also become a potential water resource for nearby communities, particularly in areas with limited clean-water availability (Bojarczuk, 2025; Opyrchał, Bąk and Romanowski, 2025; Zokirov *et al.*, 2025).

Despite its current use, the water quality of this pit lake has never been systematically evaluated. Therefore, this study aims to assess the water quality and pollution status of the Mekar Jaya pit lake using regulatory standards, STORET (STORage and RETrieval), contamination factor (CF), and pollution load index (PLI), and to examine whether water depth influences key water-quality parameters. This approach provides site-specific evidence to

support safer and more sustainable use of pit lake water by the local community.

This study aims to assess the water quality of the former mining lake in Mekar Jaya Village using the STORET, *CF*, and *PLI* methods. The parameters analysed included temperature, total suspended solids (TSS), total dissolved solids (TDS), pH, chemical oxygen demand (*COD*), dissolved oxygen (*DO*), biological oxygen demand (*BOD*), Fe, Pb, and Mn. Meanwhile, the effect of depth on water parameters was statistically tested. This approach provides a more comprehensive picture of water quality and its potential for pollution.

MATERIALS AND METHODS

RESEARCH LOCATION

This research was conducted at a former coal mine pit lake located in Mekar Jaya Village, Angsana District, Tanah Bumbu Regency, South Kalimantan Province. Geographically, Mekar Jaya Village is about 180 km from Banjarbaru City and can be reached in about 3–4 h by land. The study area is located in the Barito Sub-Basin, a sedimentary basin in South Kalimantan containing coal-bearing formations, including the Warukin Formation. The Warukin Formation comprises interbedded coal seams, sandstone, mudstone, and claystone. This geological setting is important in former mining areas because the exposure of coal-bearing rocks and sulphide minerals may promote acid mine drainage formation and metal release, thereby affecting pit lake water quality.

SAMPLING

Water samples were collected using the grab sampling method, which involves collecting water directly from the water surface at a specific time. Water sampling was conducted at six evenly distributed locations throughout the study area to provide a comprehensive representation of water quality conditions. At each sampling station, water samples were collected at two depths: 0 m, representing the surface layer of the water, and 5 m, reflecting conditions below the surface (Fig. 1).

SAMPLE PREPARATION

The water sample was then transferred into a clean sample bottle that had been rinsed with the water. Samples for *BOD* and *COD* analysis were stored at 4°C, whereas samples for metal analysis were preserved by acidification with nitric acid (HNO_3) to a pH < 2. All samples were then transported to the laboratory in insulated boxes within the holding times specified for each parameter.

RESEARCH VARIABLES AND PARAMETERS

The study evaluated pit lake water quality using the following parameters: temperature (°C), total suspended solids (TSS) ($\text{mg}\cdot\text{dm}^{-3}$), total dissolved solids (TDS) ($\text{mg}\cdot\text{dm}^{-3}$), potential of hydrogen (pH), chemical oxygen demand (*COD*) ($\text{mg}\cdot\text{dm}^{-3}$), iron (Fe) ($\text{mg}\cdot\text{dm}^{-3}$), dissolved oxygen (*DO*) ($\text{mg}\cdot\text{dm}^{-3}$), lead (Pb) ($\text{mg}\cdot\text{dm}^{-3}$), manganese (Mn) ($\text{mg}\cdot\text{dm}^{-3}$), biological oxygen demand (*BOD*) ($\text{mg}\cdot\text{dm}^{-3}$). All parameters were analysed in the

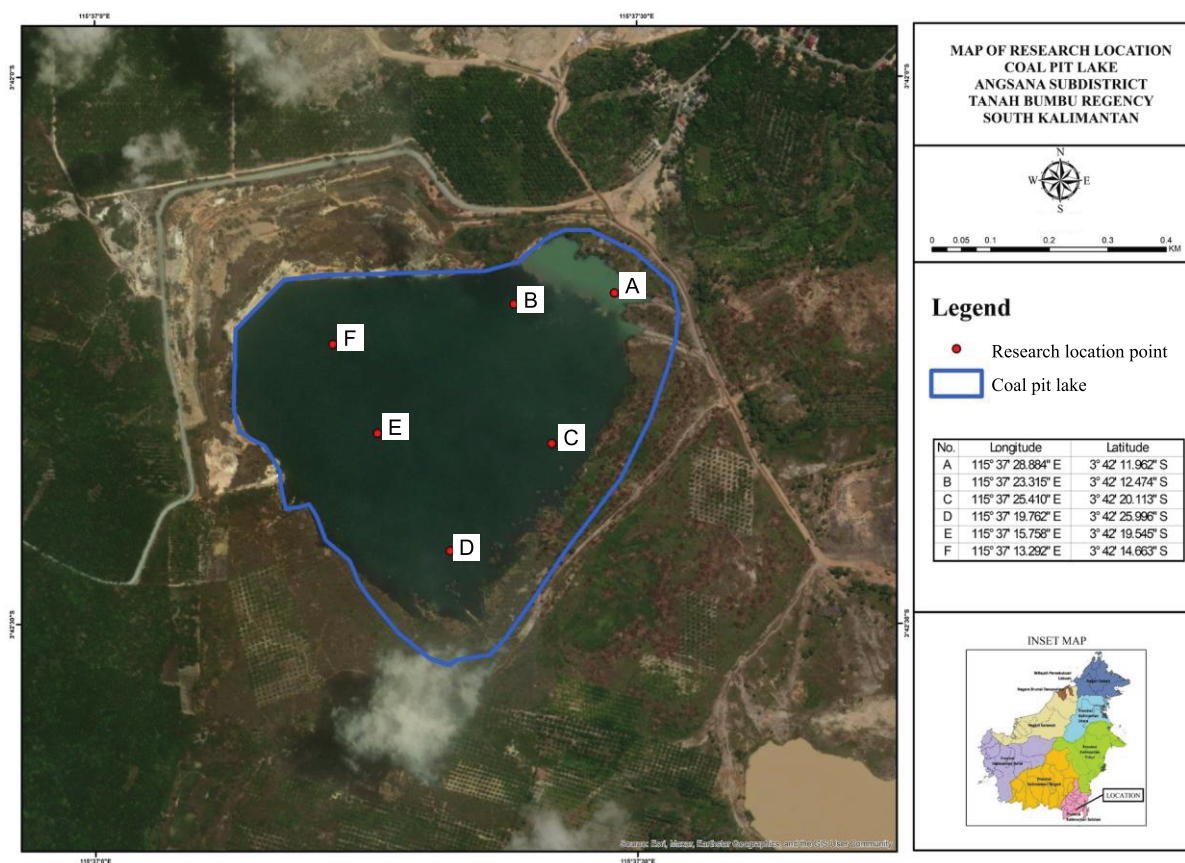


Fig. 1. Study area in Mekar Jaya Village, Angsana District, Tanah Bumbu Regency, South Kalimantan Province, Indonesia; source: Google Earth Pro imagery, modified by the authors with additional sampling points

laboratory using standard methods specified in the applicable standard methods for the examination of water and wastewater or the relevant Indonesian National Standard (BSN, 2019). Temperature, TDS, pH, DO were measured in situ using a multi-parameter water quality meter, whereas TSS was determined gravimetrically, *COD* by closed reflux spectrophotometric method, *BOD*₅ by the Winkler method, and Fe, Pb, and Mn by atomic absorption spectrophotometry (AAS). The selected parameters represent the main water-quality characteristic relevant to coal mine pit lakes, including physical condition, acidity, oxygen balance, organic pollution, and metal contamination. This selection also aligns with Indonesian Government Regulation Number 22 of 2021 (Regulation, 2021) and potential uses of pit lake water by the surrounding community. Specifically, temperature, TSS, and TDS represent physical characteristics; pH, Fe, and Mn indicate acid mine drainage-related and geochemical processes; DO, *BOD*, and *COD* reflect oxygen dynamics and organic loading; and Pb represents a toxic metal of environmental and health concern.

WATER QUALITY ANALYSIS

The data from laboratory measurements were analysed using four complementary approaches: 1) comparison with quality standards, 2) STORET method, 3) contamination factor (*CF*) and pollution load index (*PLI*), and 4) statistical test. Comparison with quality standards: the average data from 12 samples were compared with the water quality standards for classes I, II, III, and IV as outlined in Government Regulation Number 22 of 2021 concerning the implementation of environmental protection and management (Regulation, 2021). The STORET, *CF*, and *PLI* method tests use class II water quality standards, as specified in the mentioned above Government Regulation, based on their designation for the community around the current pit lake.

STORET method: this method categorises water quality into four classes based on deviations from the established quality standards:

- class A (eligible): 0,
- class B (light pollutant): between –1 and –10,
- class C (medium pollutant): between –11 and –30,
- class D (heavy pollutant): ≤ –31.

Each parameter that meets the quality standard is assigned a score of 0, and a parameter that exceeds the quality standard is assigned a negative score (score: –1) (Canter, 1977; Decree, 2003). The value system used to assess water quality status is presented in Table 1.

Contamination factor (*CF*) is calculated for each parameter using Equation (1):

$$CF = \frac{Si}{Ci} \quad (1)$$

where: *Si* = contaminant concentration (mg·dm^{–3}), *Ci* = quality standard value (research using class II water quality standards according to the Indonesian Government Regulation Number 22 of 2021).

For DO, the calculation was reversed because the water quality standard is defined as a minimum permissible concentration. This method categorises water quality into four pollution categories: low contamination: *CF* < 1, moderate contamination:

Table 1. Values according to the STORET method

Number of parameters	Value	Parameter		
		physical	chemical	biological
<10	maximum	–1	–2	–3
	minimum	–1	–2	–3
	mean	–3	–6	–9
≥10	maximum	–2	–4	–6
	minimum	–2	–4	–6
	mean	–6	–12	–18

Source: Keputusan (2003).

$1 \leq CF < 3$, considerable contamination: $3 \leq CF < 6$, and very high contamination: $CF \geq 6$.

Pollution load index (*PLI*) is calculated by Equation (2):

$$PLI = (CF_1 \cdot CF_2 \cdot CF_3 \cdot \dots \cdot CF_n)^{\frac{1}{n}} \quad (2)$$

where: *CF*₁, *CF*₂, *CF*₃, *CF*_{*n*} = contamination factors of each analysed metal or pollutant, *n* = the number of parameters analysed.

The *PLI* was used to evaluate the overall pollution levels resulting from multiple contaminants. Water quality was classified into three categories: *PLI* = 1, baseline pollution level, *PLI* > 1, polluted, *PLI* < 1, unpolluted. The pit lake water quality and its suitability for community use were assessed based on quality standard comparisons and pollution level analyses (STORET, *CF*, *PLI*). The potential to improve water quality through treatment is also evaluated based on parameters that do not meet quality standards.

STATISTICAL TESTS

Water samples were collected at two depths, namely 0 m and 5 m. The 0 m depth samples represented the surface layer directly influenced by atmospheric exchange and the water most accessible for direct community use. In contrast, the 5 m depth samples represented subsurface conditions and were used to assess possible vertical variation in water quality within the pit lake. This depth-based approach was intended to identify whether key physicochemical parameters differed between the surface and lower water layer. The pollution status was evaluated using STORET, *CF*, and *PLI*, while differences between depths were assessed using appropriate statistical tests based on data characteristics. Symbols, units, and numerical presentation were standardised consistently throughout the manuscript. Data analysis was conducted to determine differences in water quality at two sampling depths: 0 m and 5 m. The evaluation included descriptive analysis, whereas assumption testing and differential tests were used to determine the significance of differences between depths. Raw data on temperature parameters, TSS, TDS, pH, *COD*, *BOD*, DO, Fe, Mn, and Pb were first entered into Statistical Package for the Social Sciences (SPSS). The results of the descriptive analysis were used to examine the tendency of the middle value and the distribution of the data, as a basis for the initial interpretation of the distribution of water quality parameters.

Because the number of observations at each sampling depth was limited, the statistical analyses were intended as an exploratory assessment of potential depth-related differences rather than as a basis for broad generalisation. The independent samples *t*-test was applied to variables that satisfied the assumptions of normality, whereas the Mann-Whitney *U* test was used for data that did not meet the assumptions of parametric testing. The results were interpreted cautiously, recognising that the limited sample size reduced statistical power and may have constrained the ability to detect subtle differences between depths.

RESULTS AND DISCUSSION

WATER QUALITY OF PIT LAKE

The results of laboratory analysis of 12 water samples (6 samples at a depth of 0 m and six samples at a depth of 5 m) from a coal pit in Mekar Jaya Village are presented in Table 2.

As shown in Table 2, all pit lake water parameters measured meet Class IV water quality standards. When assessed against the

Class I water quality standards, the pit lake water exceeded the permissible limits of TSS, *COD*, and *BOD*. If the pit lake water is compared to the class II water quality standards, it does not meet the *BOD* parameter. This condition shows that the primary contaminant in the pit lake water is organic. Similar findings have been reported in previous studies, which found that lake water contaminants include ammonia, iron, manganese, organic matter, and total coliforms (Zhou *et al.*, 2020; Fuentes-López *et al.*, 2022).

WATER POLLUTION STATUS OF PIT LAKE

To comprehensively assess the pollution status of the pit lake water. Three analysis methods were used: STORET, *CF*, and *PLI*.

STORET method

The water quality status was evaluated using the STORET method based on the Class II water quality standards, reflecting the current uses of the pit lake for bathing, washing, livestock, plantations, and fish farming. As shown in Table 3, *BOD* exceeded the Class II standard, resulting in STORET scores of -4

Table 2. Coal mine pit lake water quality measurement results

Sample	Parameter									
	T (°C)	pH	TSS	TDS	DO	COD	BOD	Fe	Pb	Mn
			mg·dm ⁻³							
A0	25.9	6.99	27	42	6.6	14.33	6.31	0.154	<0.004	<0.05
A5	25.9	6.98	33	34	6.3	16.84	6.31	0.249	<0.004	<0.05
B0	26.7	7.27	22	34	6.2	14.33	6.31	0.167	<0.004	<0.05
B5	27.2	6.96	42	34	6.2	16.13	9.01	0.156	<0.004	<0.05
C0	26.7	7.21	35	35	6.6	15.05	7.21	0.156	<0.004	<0.05
C5	27.1	6.95	31	35	6.5	16.84	9.01	0.146	<0.004	<0.05
D0	26.7	7.16	25	34	6.5	16.48	7.21	0.262	<0.004	<0.05
D5	26.1	6.86	43	35	6.2	16.84	8.11	0.165	<0.004	<0.05
E0	26.7	7.1	26	34	6.4	16.84	9.01	0.190	<0.004	<0.05
E5	27.7	6.86	23	35	6.9	14.33	9.01	0.152	<0.004	<0.05
F0	26.7	7.09	28	34	6.3	16.84	9.91	0.125	<0.004	<0.05
F5	27.6	6.86	36	38	6.4	16.84	9.01	0.150	<0.004	<0.05
Max.	27.2	7.27	43	42	6.9	16.84	9.01	0.262	<0.004	<0.05
Min.	25.9	6.86	22	34	6.2	14.33	6.31	0.125	<0.004	<0.05
Mean	26.75	7.02	30.92	35.33	6.43	15.97	8.04	0.173	<0.004	<0.05
Class I quality standards ¹⁾										
Deviation ±3 from natural T	6–9		25	1000	6	10	2	0.3	0.03	0.4
Class II quality standards ¹⁾										
Deviation ±3 from natural T	6–9		50	1000	4	25	3	–	0.03	0.4
Class III quality standards ¹⁾										
Deviation ±3 from natural T	6–9		100	1000	3	40	6	–	0.03	0.5
Class IV quality standards ¹⁾										
Deviation ±3 from natural T	6–9		400	1000	1	80	12	–	0.5	1.0

¹⁾ According to Regulation (2021).

Explanations: *T* = temperature, pH = potential of hydrogen, TSS = total suspended solids, TDS = total dissolved solids, DO = dissolved oxygen, *COD* = chemical oxygen demand, *BOD* = biological oxygen demand.

Source: own study.

Table 3. Water quality status test using the STORET method

Parameter	Unit	Class II quality standard	Measurement result			Score
			max.	min.	mean	
<i>T</i>	°C	deviation ± 3 from natural <i>T</i>	27.2	25.9	26.75	0
pH	–	6–9	7.27	6.86	7.02	0
TSS	mg·dm ⁻³	50	43	22	30.92	0
TDS		1,000	42	34	35.33	0
DO		4	6.9	6.2	6.43	0
<i>COD</i>		25	16.84	14.33	15.97	0
<i>BOD</i>		3	9.01	6.31	8.04	–20
Fe		–	0.262	0.125	0.173	0
Pb		0.03	<0.004	<0.004	<0.004	0
Mn		0.4	<0.05	<0.05	<0.05	0

Explanations as in Tab. 2.

Source: own study.

for maximum value, –4 for the minimum value, and –12 for the mean value. The total STORET score was –20, corresponding to Class C (moderately polluted). This condition indicates a negative impact on water quality from previous mining activities, particularly from organic matter. Contamination is entirely due to high *BOD* at all sampling points. These results indicate that organic pollution was the principal factor contributing to the moderate pollution status of the pit lake.

Contamination factor

The *CF* method indicates that, in general, water quality remains good, as almost all parameters have *CF* < 1 (Tab. 4). One parameter indicates pollution, i.e. *BOD* (*CF* = 2.68), indicating moderate levels of contamination. This condition shows that the input of organic pollutants is relatively high, as for the other parameters (TSS, TDS, *COD*, DO, Fe, Mn, Pb), they remain at low levels of contamination (safe).

Pollution load index

Based on Equation (2) and Table 3, the pit lake had a *PLI* value of 0.348, indicating a low overall pollution load. This result suggests that, when all measured parameters are considered collectively, the pit lake water is not heavily polluted. However, the *PLI* should not be interpreted in isolation. The *CF* results indicate that *BOD* was the only parameter exhibiting a moderate level of contamination (*CF* = 2.68). In contrast, TSS, TDS, *COD*, DO, Fe, Pb and Mn remained in the low contamination category (*CF* < 1). This pattern indicates that the main water-quality concern is not widespread multi-parameter contamination but rather a localised organic pollution signal reflected by the elevated *BOD* concentration.

The combined interpretation of STORET, *CF*, and *PLI* provides a more comprehensive assessment of pit lake water quality. The STORET evaluates water quality against regulatory standards and is therefore useful for determining compliance with designated water-use classes. The *CF* identifies the parameter contributing most strongly to the contamination profile and, in

Table 4. Water contamination test using the contamination factor (*CF*) method

Parameter	<i>Si</i>	<i>Ci</i>	<i>CF</i>	Contamination category
TSS	30.92	50	0.62	low
TDS	35.33	100	0.035	low
DO	6.43	4	0.62	low
<i>COD</i>	15.97	25	0.64	low
<i>BOD</i>	8.04	3	2.68	medium
Fe	0.173	0.3	0.58	low
Pb	<0.004	0.03	0.13	low
Mn	<0.05	0.4	0.125	low

Explanations: *Si* = contaminant concentration (mg·dm⁻³), *Ci* = quality standard value (research using class II water quality standards according to the Regulation (2021)), the other as in Tab. 2.

Source: own study.

this study, identified *BOD* as the principal contributor. The *PLI* summarises the cumulative pollution burden and shows that the lake is not under severe overall pressure. Taken together, these results suggest that the pit lake is generally in relatively good condition, although its suitability for community use is limited by elevated organic pollution, as reflected by *BOD* rather than by widespread chemical or heavy-metal contamination. Therefore, management efforts should focus primarily on reducing organic matter inputs and *BOD* concentrations while maintaining continued monitoring of other water quality parameters.

POTENTIAL USE OF WATER PIT LAKE

Pit lake water may provide valuable water resources for surrounding communities, including flood protection, hydro-power generation, agricultural land irrigation, and water storage

for community consumption (Opyrchał, Bąk and Romanowski, 2025; Panigrahi *et al.*, 2025). The pit lake in Mekar Jaya village has sufficient water quantity to meet local demand. However, in terms of quality, it is necessary to study its use management (Zokirov *et al.*, 2025). The use of pit lake water is based on Government Regulation No. 22 of 2021 (Regulation, 2021). Parameters that exceed Class I water quality standards (TSS, *COD*, and *BOD*) indicate the presence of organic matter can interfere with drinking water treatment and have the potential to form harmful by-products. Pit lake water is also not suitable for recreational infrastructure/facilities, or freshwater fish farming, animal husbandry and irrigation (class II water) because the *BOD* value ($8.04 \text{ mg}\cdot\text{dm}^{-3}$) exceeds the maximum limit of $6 \text{ mg}\cdot\text{dm}^{-3}$ (Tab. 2). However, DO levels remained between 6.2 and $6.9 \text{ mg}\cdot\text{dm}^{-3}$ during sampling. Therefore, the relationship between *BOD* and DO decline in this study does not support a direct causal relationship. The measured *BOD* values are likely influenced by organic matter input from domestic activities, surface runoff, aquaculture activities, and vegetation decomposition. However, these potential sources remain hypothetical, based on local environmental conditions, and have not been directly investigated in this study. Based on the measured water quality, the pit lake complied with the Class III water quality standards, indicating its suitability for freshwater fish farming, animal husbandry, and irrigation.

Although currently only suitable for class III, the water pit lake in Mekarjaya village has strategic potential for improvement towards compliance with the Class I and II water quality standards. This assessment is supported by the fact that basic parameters, such as heavy metals (Pb, Mn, Fe) and pH, already comply with the more stringent Class I and II requirements. The principal limitation is the elevated *BOD* concentration. The quality of pit lake water in the village of Mekarjaya is better than that of the pit lake water in North-eastern Mexico, Poland, and Marocco (Habchaoui *et al.*, 2024; Kemanga, McIntyre and Bulovic, 2024; Kruk *et al.*, 2024; Batista-Cruz *et al.*, 2025; Kaznowska, Wasilewicz and Barszcz, 2026).

Several alternative technologies have been developed to improve the quality of polluted water (Edwin *et al.*, 2025; Jucherski *et al.*, 2025). Integrated approaches combining biological and physicochemical methods have also been integrated under this approach. In addition, adsorption has been applied to remove organic and inorganic compounds, including phosphates, ammonium, nitrates, and heavy metals (Kusdarini and Budianto, 2022; Budianto *et al.*, 2023; Kusdarini, Sania and Budianto, 2023; France *et al.*, 2024; Edwin *et al.*, 2025; Kusdarini *et al.*, 2025a; Kusdarini *et al.*, 2025b). By applying appropriate and cost-effective treatment technologies, pit lake water, which initially suffered due to the environmental impact of mining activities, can be transformed into a valuable water resource to meet community clean water needs, especially as a source of raw water for drinking (Major *et al.*, 2024; Iranmanesh *et al.*, 2025; Li *et al.*, 2025).

STATISTICAL TESTS

Data normality test

The Saphiro–Wilk normality test in this study was used to assess whether the data were normally distributed (Tab. 5).

Most water quality parameters at both sampling depths met the assumption of normality. However, some parameters are not normally distributed. Therefore, the effects of depth on TSS, pH, DO, Mn, and Pb was analysed using the parametric independent samples *t*-test because the data at both depths were normally distributed. In contrast, temperature, TDS, *COD*, *BOD*, and Fe were analysed using the nonparametric test with the Mann–Whitney *U* test because these data did not satisfy the assumption of normality.

Data homogeneity test

To determine whether data are distributed similarly across groups, this study used Levene's test (Tab. 6).

The results of the homogeneity test are presented in Table 6. Most parameters, including TSS, TDS, pH, *COD*, *BOD*, DO, Fe,

Table 5. Data of Saphiro–Wilk normality test

Parameter	Depth from the surface of the water (m)	Statistic	<i>df</i>	Sig.	Normality
Temperature (°C)	0	0.552	5	0.000	abnormal
	5	0.906	5	0.441	normal
TSS ($\text{mg}\cdot\text{dm}^{-3}$)	0	0.885	5	0.335	normal
	5	0.924	5	0.557	normal
TDS ($\text{mg}\cdot\text{dm}^{-3}$)	0	0.633	5	0.002	abnormal
	5	0.684	5	0.006	abnormal
pH	0	0.980	5	0.937	normal
	5	0.813	5	0.103	normal
<i>COD</i> ($\text{mg}\cdot\text{dm}^{-3}$)	0	0.846	5	0.183	normal
	5	0.715	5	0.014	abnormal
<i>BOD</i> ($\text{mg}\cdot\text{dm}^{-3}$)	0	0.833	5	0.146	normal
	5	0.735	5	0.021	abnormal

cont. Tab. 5

Parameter	Depth from the surface of the water (m)	Statistic	df	Sig.	Normality
DO (mg·dm ⁻³)	0	0.881	5	0.314	normal
	5	0.831	5	0.141	normal
Fe (mg·dm ⁻³)	0	0.792	5	0.070	normal
	5	0.702	5	0.010	abnormal
Mn (mg·dm ⁻³)	0	0.990	5	0.980	normal
	5	0.901	5	0.415	normal
Pb (mg·dm ⁻³)	0	0.852	5	0.200	normal
	5	0.965	5	0.841	normal

Explanations: *df* = degree of freedom, Sig. = significance, the other as in Tab. 2.

Source: own study.

Table 6. Data of homogeneity test

Parameter	Statistic	Sig.	Homogeneity
Temperature (°C)	5.942	0.041	not homogeneous
TSS (mg·dm ⁻³)	1.991	0.196	homogeneous
TDS (mg·dm ⁻³)	4.421	0.069	homogeneous
pH	1.347	0.279	homogeneous
COD (mg·dm ⁻³)	0.449	0.522	homogeneous
BOD (mg·dm ⁻³)	0.103	0.757	homogeneous
DO (mg·dm ⁻³)	1.449	0.263	homogeneous
Fe (mg·dm ⁻³)	0.014	0.908	homogeneous
Mn (mg·dm ⁻³)	0.442	0.525	homogeneous
Pb (mg·dm ⁻³)	18.448	0.003	not homogeneous

Explanations: Sig. = significance, other as in Tab. 2.

Source: own study.

and Mn, showed significance values greater than 0.05, whereas temperature and Pb had significance values below 0.05. Therefore, the data variances across sampling depths can be considered homogeneous for most parameters.

Meanwhile, only two parameters, namely temperature and Pb, showed significance values below 0.05. Therefore, the variances between sampling depths for these parameters were considered heterogeneous.

Test the effect of water surface depth on water quality parameters

Testing the effect of water surface depth on water quality parameters was performed using an independent *t*-test or a Mann–Whitney *U* test (Tab. 7).

Based on the results of the independent-samples *t*-test (a) and Mann–Whitney *U* test (b) comparing the 0 m and 5 m sampling depths (Tab. 7), only pH showed a statistically significant difference between depths (*p* = 0.003). The value is less than 0.05, so it can be concluded that there is a significant

Table 7. Test the effect of the depth of the water surface on water quality parameters

Variable	Value at depth (mean ±SD)		Δ mean	<i>t</i> statistics	<i>p</i> -value
	0 m	5 m			
Temperature	26.54 ±0.36	26.80 ±0.77	−0.26000	−0.865b	0.387
TSS	27.00 ±4.85	34.40 ±8.29	−7.40000	−1.722a	0.123
TDS	35.80 ±3.49	34.60 ±0.55	1.20000	−0.231b	0.817
pH	7.15 ±0.11	6.92 ±0.06	0.22400	4.110a	0.003
COD	15.41 ±1.19	16.20 ±1.09	−0.79000	−1.092b	0.275
BOD	7.21 ±1.10	8.29 ±1.17	−1.08000	−1.315b	0.189
DO	6.46 ±0.17	6.42 ±0.29	0.04000	0.264a	0.799
Fe	0.19 ±0.04	0.17 ±0.04	0.01220	−1.048b	0.295
Mn	0.03 ±0.02	0.02 ±0.02	0.0042248	−0.522a	0.602
Pb	0.002 ±0.002	0.001 ±0.001	0.0003850	0.481a	0.652

Explanations: *SD* = standard deviation, “a” = the results obtained under the assumption of equal variances (equal variance assumed), “b” = the results obtained when equal variances were not assumed (equal variance not assumed), the other as in Tab. 2.

Source: own study.

difference in pH between 0 m (7.15 ± 0.11) and 5 m (6.92 ± 0.06). This condition suggests that the water in the surface layer is slightly more alkaline than that in the deeper layer. In contrast, temperature, TSS, TDS, COD, BOD, DO, Fe, Mn, and Pb showed no significant differences between the two sampling depths (all $p > 0.05$).

These results show that the water quality profile was relatively homogeneous from the surface to a depth of 5 m, with pH being the only parameter showing noticeable difference. Overall, the physical and chemical characteristics of the water were vertically stable, although a slight decrease in pH was observed at the deeper sampling depth.

The absence of statistically significant differences for most parameters should therefore be interpreted with caution. Given the limited number of observations, these results do not necessarily indicate the complete absence of vertical variation but rather that no statistically significant differences were detected within the present dataset.

CONCLUSIONS

This study demonstrated that the Mekar Jaya pit lake had a low overall pollution load based on the combined interpretation of the STORET (STOrage and RETrieval), contamination factor (CF), and pollution load index (PLI). However, biological oxygen demand (BOD) indicated moderate organic contamination and was the main water quality concern. Water quality was generally stable from the surface to a depth of 5 m, with pH being the only parameter that differed significantly between depths, indicating limited vertical variation within the sampled water column. These findings suggest that the pit lake has the potential to provide a reliable water resource. However, its current water quality still requires treatment before wider community use, particularly for higher-quality water-use classes. Therefore, management efforts should prioritise reducing organic pollution while maintaining regular water quality monitoring to support safe and sustainable utilisation. Further studies should evaluate the technical feasibility of specific end uses, including irrigation, livestock, fisheries, clean water, and drinking-water supply. This research contributes to Sustainable Development Goal (SDG) 6, namely, clean water and proper sanitation. The potential use of water for agriculture, livestock, or fisheries also contributes to SDGs 1 (No Poverty), 2 (Zero Hunger), and 3 (Good health and Well-being).

CONFLICT OF INTERESTS

All authors declare that they have no conflicts of interest.

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